Code-Layout, Readability and Reusability

Date	5 july 2026
Team ID	
Project Name	HDI Prediction System using Machine Learning and Flask
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing and proper indentation are maintained throughout the code.	Yes	Proper indentation maintained throughout the project.
2	Proper File Structure	Files and folders are logically organized into appropriate directories.	Yes	Project structure is well organized.
3	Meaningful Variable Names	Variables are named clearly to reflect their purpose and functionality.	Yes	Descriptive variable names are used.
4	Function / Method Names	Functions are named according to their specific tasks and operations.	Yes	Function names improve readability.
5	Code Comments	Important sections include comments to explain the program logic.	Partial	Comments are provided where necessary.
6	Modular Design	Code is divided into reusable functions and modules for better maintenance.	Yes	Modular approach improves reusability.
7	No Redundant Code	Duplicate or unnecessary code is removed to improve efficiency.	Yes	Code duplication has been minimized.
8	Error Handling	Exceptions and invalid inputs are handled appropriately to avoid failures.	Yes	Basic exception handling is implemented.

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
1	Data Preprocessing	Python, Pandas	Model Training	High
2	Machine Learning Model	Scikit-learn	Prediction Module	High
3	Flask Backend	Flask	Web Application	High
4	HTML Templates	HTML, CSS	User Interface	High
5	Prediction Function	Python	App Backend	High

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Well-organized folder structure and modular code.
Readability	5	Clear variable names and simple logic.
Reusability	5	Components are reusable across the application.
Documentation / Comments	4	Important sections are documented with comments.
Overall Score	5/5	Clean, maintainable, and reusable project code.